

# RESEARCH ROADMAP

## A Graph Reinforcement Learning Framework for Routing in a Hybrid Fiber and Free-Space QKD Network

### Agent-executable research, experiment, validation, and paper-writing plan

Visual system matched to the uploaded QSMS Presentation Script: white background, DejaVu Sans typography, burgundy #9E1735 headings/lines, pale burgundy #F6F1F2 panels, dark #242424 body text.

This document is intentionally operational. It tells a research agent what is already implemented, what evidence exists, what is still missing, which dependency unlocks each task, where each equation/reference belongs, and what figure/table should be produced from each experiment.

### Source set

1. Current draft: QKD\_IEEE\_Paper.pdf. 2. Base paper: Johann et al., JOCN 2024, Comparison and optimization of different routing methods for meshed QKD networks using trusted nodes. 3. Uploaded QSMS script used only as the visual/font/color reference for this roadmap.

## 1. STATUS LEGEND — HOW AN AGENT SHOULD READ THIS

| STATUS          | Meaning                                                                                                               | Agent action                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>DONE</b>     | Implemented and described in the current draft, or supported by an archived result.                                   | Do not rebuild from scratch. Verify reproducibility and improve presentation/validation if marked elsewhere. |
| <b>PARTIAL</b>  | Implementation exists but evidence is incomplete, single-condition, assumption-heavy, or not independently validated. | Treat as existing infrastructure; prioritize validation before making comparative claims.                    |
| <b>ARCHIVED</b> | A result/configuration exists, but it is not sufficient as final evidence.                                            | Preserve it as provenance. Regenerate under the corrected protocol before using it as a main result.         |
| <b>TODO</b>     | Not implemented or not measured.                                                                                      | Build only after prerequisites are satisfied.                                                                |
| <b>BLOCKED</b>  | Cannot be meaningfully completed until another model/data layer exists.                                               | Do not fake a result. Complete the dependency first.                                                         |

**Non-negotiable rule: a simulator quantity is not a measurement simply because it has a numerical value. The current paper explicitly distinguishes modeled outputs, assumptions, calibration evidence, and field validation.**

## 2. CURRENT STATE — WHAT IS ALREADY DONE

| Workstream                        | Status          | Evidence / current implementation                                                                                                                                                                                                         | Final-paper use                                                      |
|-----------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Hybrid topology                   | <b>DONE</b>     | 7 Indian city hubs; 2,370 modeled nodes; 2,653 links; 273 fiber + 2,380 candidate FSO. Road geometry is a routing proxy.                                                                                                                  | Methods + topology figure; explicitly label as proxy.                |
| Fiber model                       | <b>PARTIAL</b>  | 1550 nm; 0.20 dB/km; detector efficiency 0.93; dark-count bound $<70 \text{ s}^{-1}$ ; excess loss included.                                                                                                                              | Methods + analytical validation; provenance table.                   |
| FSO model                         | <b>PARTIAL</b>  | 785 nm scenario; log-normal correlated turbulence; atmospheric/diffraction/pointing/background terms; detector noise. No Gamma-Gamma or generalized amplitude damping.                                                                    | Methods + stress experiments; assumptions must remain visible.       |
| Rate model                        | <b>PARTIAL</b>  | $r_{\text{proxy}} = p_{\text{det}}[1 - 2h_2(Q)]$ ; normalized simulator utility, not bps and not a composable security certificate.                                                                                                       | Methods Eq. (1) + explicit scope limitation.                         |
| QBER feasibility                  | <b>DONE</b>     | Hard $Q_{\text{max}} = 0.11$ action mask; projected chain-error constraint also used.                                                                                                                                                     | Methods + correctness tests.                                         |
| Observation                       | <b>DONE</b>     | 5 node features + 9 edge features, including QBER, rate proxy, FSO flag, distance, projected error, pool state, switching, visited flag, progress.                                                                                        | Methods Table II + ablation.                                         |
| GNN-PPO                           | <b>DONE</b>     | 3 message-passing layers; shared actor-critic trunk; masked action head; PPO; optional attention.                                                                                                                                         | Methods + architecture figure + ablation.                            |
| Classical baselines               | <b>PARTIAL</b>  | Random feasible, BFS, Dijkstra, Max-SKR/rate selector exist; old baseline outputs invalid because selector did not consistently honor mask.                                                                                               | Regenerate all under one corrected protocol.                         |
| Archived GNN result               | <b>ARCHIVED</b> | Mumbai→Kolkata, monsoon, $22:00 \pm 1 \text{ h}$ , 200 epochs resumed; 10/10 success; mean 25.6 hops.                                                                                                                                     | Use only as preliminary/single-condition evidence.                   |
| Shortest-path reference           | <b>DONE</b>     | Independent NetworkX check: 24-edge unweighted shortest path; old 23-edge reference was incorrect.                                                                                                                                        | Correct the paper and all plots/captions.                            |
| Controlled link-choice diagnostic | <b>ARCHIVED</b> | 384 paired comparisons; FSO higher proxy in all sampled pairs; trained policy selected FSO zero times; action probability near 1/2.                                                                                                       | Main diagnostic showing current policy weakness; motivate ablations. |
| Physics evidence                  | <b>PARTIAL</b>  | 80-km fiber: 16.46 dB modeled loss, $p_{\text{det}} \approx 0.021$ , $\text{QBER} \approx 0.0105$ , $\text{proxy} \approx 0.0175$ . 10-km FSO: modeled availability 88% config / 88.5% in 1,000 samples; median conditional proxy 0.2035. | Report as modeled outputs, not measured performance.                 |
| ILP                               | <b>PARTIAL</b>  | Mathematical comparator Eq. (5)–(8) already written in current draft; not executable/measured because demand/QKP units are absent.                                                                                                        | Keep as future/global reference until demand layer exists.           |

**Current evidence is therefore strongest at the simulator + route-controller level, not yet at the sustained network-service level.**

### 3. WHAT IS STILL MISSING — IN DEPENDENCY ORDER

| Priority | Missing layer                          | Dependency                             | Why it matters                                                | Paper consequence                                               |
|----------|----------------------------------------|----------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| P0       | Corrected baseline regeneration        | Gymnasium/runtime + common mask        | All comparative claims currently depend on this.              | Replace old baseline table; add statistically valid comparison. |
| P0       | 5-seed training/evaluation             | Stable evaluation runtime              | One archived run cannot establish policy behavior.            | Report mean/spread and seed sensitivity.                        |
| P0       | Physics/noise analytical validation    | Independent calculations / code checks | Separates model correctness from RL behavior.                 | Strengthens Methods + Validation.                               |
| P0       | Controlled link-choice/QBER/FSO stress | Existing simulator                     | Tests whether the policy actually uses physical features.     | Core mechanism result.                                          |
| P1       | Endpoint-pair generalization           | Multiple endpoint runs                 | Current result is one pair.                                   | Generalization subsection.                                      |
| P1       | Season/diurnal generalization          | Scenario generator                     | Current result is one monsoon-night condition.                | Robustness subsection.                                          |
| P1       | Feature/architecture/reward ablations  | Training pipeline                      | Current policy weakness needs attribution.                    | Ablation table/figures.                                         |
| P1       | Inference benchmark                    | Runtime harness                        | Current paper has no measured target-hardware latency.        | Deployment/complexity subsection.                               |
| P1       | Exact small-graph oracle               | Tiny graph generator                   | Provides route-quality reference without ILP scope explosion. | Oracle/regret result.                                           |
| P2       | Absolute QKP/key resource model        | Absolute key-rate model                | Normalized pool cannot support network-level service metrics. | Unlocks Johann-style experiments.                               |
| P2       | Demand matrix + temporal demand        | Trace or declared synthetic model      | Needed for blocking/load balancing/prediction.                | Unlocks network-service study.                                  |
| P2       | Key consumption + refill               | Absolute rate + QKP capacity           | Creates temporal resource coupling.                           | Network-operation on simulator.                                 |
| P2       | Blocking/load balancing                | Demand + QKP loop                      | Directly comparable to Johann KPIs.                           | Journal-level result.                                           |
| P2       | Management traffic                     | Controller/query protocol definition   | Cannot measure overhead without message semantics.            | Security/overhead result.                                       |
| P2       | Weight optimization                    | QKP state + demand loop                | Classical state-aware comparator.                             | Journal baseline.                                               |
| P2       | LSTM demand prediction                 | Historical demand matrices             | Only meaningful with dynamic demand.                          | Journal temporal comparator.                                    |

| Priority | Missing layer                 | Dependency                                          | Why it matters                                      | Paper consequence              |
|----------|-------------------------------|-----------------------------------------------------|-----------------------------------------------------|--------------------------------|
| P2       | Perfect prediction            | Future demand matrix                                | Oracle upper reference for prediction methods.      | Journal reference.             |
| P2       | ILP execution                 | Demand matrix + absolute capacity + candidate paths | Current ILP is only a formulation.                  | Global demand-level reference. |
| P3       | Measured traces / HIL / field | Calibrated hardware/channel data                    | External validation only after simulator is stable. | Validation/limitations.        |

## 4. P0 — IMMEDIATE PAPER EXPERIMENTS

| ID   | Experiment              | Procedure                                                                                                                                                                     | Output                                                          | Write it as                                                                 |
|------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------|
| P0-1 | Corrected baselines     | Run Random-feasible, BFS, Dijkstra, Max-SKR with identical state, endpoint, seed set, termination rule and valid-action mask.                                                 | CSV + summary table + distributions.                            | “Under a common feasibility protocol...”                                    |
| P0-2 | Physics checks          | Sweep fiber distance; independently calculate loss/detection; sweep QBER/noise; validate rate-proxy monotonicity; sample FSO availability/correlation.                        | 4–6 validation plots.                                           | “The simulator reproduces the expected limiting trends...”                  |
| P0-3 | Link-choice sensitivity | Construct matched candidate links with controlled $\Delta R$ , QBER margin, distance and outage; hold route progress constant; record action probability and selected action. | Probability-vs- $\Delta R$ scatter/curve + selection-rate plot. | “The policy does/does not respond monotonically to...”                      |
| P0-4 | QBER stress             | Evaluate QBER levels 0.01, .03, .05, .07, .09, .10, .109 and $>.11$ ; separate hard-mask behavior from preference among feasible links.                                       | Success/entropy/probability curves.                             | “Feasibility and preference are evaluated separately...”                    |
| P0-5 | FSO stress              | Vary turbulence, availability, pointing/background and outage persistence independently where supported.                                                                      | Robustness curves/heatmap.                                      | “Performance degrades/recoveries under controlled channel perturbations...” |
| P0-6 | 5-seed stability        | Train 5 independent policies with identical protected hyperparameters; fixed evaluation seeds.                                                                                | Return/success/hops distributions + training curves.            | “Across independent training seeds...”                                      |

**Stop condition for the immediate paper: do not expand to demand prediction or ILP experiments until P0 produces a trustworthy baseline comparison and establishes whether the learned policy actually uses the physical link features.**

## 5. P1 — GENERALIZATION, ABLATION, AND INFERENCE

| ID   | Work                     | Minimum design                                                                          | Primary figure/table                                  |
|------|--------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------|
| P1-1 | Endpoint generalization  | Train on selected pairs; test held-out source-destination pairs among the 7 hubs.       | Train/test matrix or held-out endpoint bar/box plot.  |
| P1-2 | Season/diurnal           | Clear/monsoon × day/night; hold out at least one condition from training.               | Condition heatmap: success, hops, proxy, outage.      |
| P1-3 | Message-passing ablation | 0/1/2/3 layers under matched parameter/training budget where possible.                  | Ablation bar/box plot.                                |
| P1-4 | Attention ablation       | Same seed set and budget, GNN vs attention.                                             | Paired distribution + action-probability diagnostic.  |
| P1-5 | Feature ablation         | Remove QBER, rate, pool, progress, channel-type features one group at a time.           | Performance delta table + tornado-style bar chart.    |
| P1-6 | Reward ablation          | Remove/perturb rate, pool, delay, energy, congestion, switching, progress terms.        | Metric matrix; avoid claiming causality from one run. |
| P1-7 | Small-graph oracle       | Enumerate feasible paths on tiny graphs and compare selected route utility/regret.      | Regret CDF / route-quality scatter.                   |
| P1-8 | Inference benchmark      | Measure model forward, decision, route, p50/p95/p99, CPU/GPU, memory; scale graph size. | Latency distribution + latency-vs-graph-size plot.    |

**Inference comparison rule: use Dijkstra as the primary classical per-decision reference and Max-SKR as a second reference. Do not put ILP on the same per-next-hop latency plot; ILP is a global demand-level optimizer.**

## 6. P2 — JOURNAL EXTENSION: THE NETWORK-SERVICE LAYER

### Target transformation

**Current paper:** channel state → route → next hop. **Journal extension:** demand → route → key consumption → QKP depletion → QKD generation/refill → next demand.

| Stage                      | Implement                                                                          | Then measure                            | Unlocks                      |
|----------------------------|------------------------------------------------------------------------------------|-----------------------------------------|------------------------------|
| J-1 Absolute key resources | Per-edge key capacity, initial key count, key bits/keys, absolute generation rate. | QKP inventory over time.                | Resource-aware routing.      |
| J-2 Demand stream          | Time-indexed source/destination/data requests; preserve trace provenance.          | Demand volume and temporal variation.   | Blocking/load balancing.     |
| J-3 Consumption            | For a selected path, consume required keys from every QKP.                         | Per-demand accepted/rejected state.     | Finite resource coupling.    |
| J-4 Refill                 | Generate/distribute keys according to link generation rate at defined intervals.   | QKP recovery curves.                    | Sustained operation.         |
| J-5 Blocking               | Reject demand if any path QKP lacks sufficient keys.                               | Blocking probability vs traffic factor. | Direct Johann KPI.           |
| J-6 Load balance           | Track mean/min/variance/filling levels per edge/region.                            | QKP utilization heatmaps/time series.   | Traffic-engineering result.  |
| J-7 Management traffic     | Define controller→QKP queries and count them per routing demand/time interval.     | Requests/bytes vs traffic.              | Information-overhead result. |

Base-paper provenance: Johann et al. describe the centralized controller/QKP architecture on p. 385, the dynamic demand/key-consumption/refill loop on p. 386, blocking on p. 389, QKP utilization/load balancing on p. 389–390, and management traffic on p. 390. Their study uses 10,000 initial keys of 256 bits per QKP and periodic generation/refill; if reproducing those assumptions, cite Johann directly rather than presenting them as your own model.

## 7. EQUATION REGISTRY — EXACTLY WHERE TO GET EACH EQUATION

| Equation / object                   | Use in paper                                   | Exact source                                                                                                                          | Agent instruction                                                                                                                                  |
|-------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Normalized rate proxy               | Current link-utility definition.               | Current draft, Section II-B, Eq. (1), PDF p. 4.                                                                                       | Copy the existing equation exactly and label it a simulator proxy. Do not call it bps or a composable key rate.                                    |
| Discounted RL objective $J(\theta)$ | Formal routing objective.                      | Current draft, Section IV-A, Eq. (2), PDF p. 5; RL foundation cited to [8], [25], [26].                                               | Retain current notation. Explain hard feasibility separately from reward optimization.                                                             |
| Masked policy $\pi_\theta(u o)$     | Hard-feasibility action selection.             | Current draft, Section IV-C-2, Eq. (3), PDF p. 5.                                                                                     | Use current equation. Explain that invalid actions are removed before softmax.                                                                     |
| PPO clipped surrogate               | Training objective.                            | Current draft, Section IV-C-3, Eq. (4), PDF p. 5; original PPO = reference [8].                                                       | Keep equation from current draft; cite Schulman et al. [8] for PPO.                                                                                |
| GNN/message passing                 | Architecture definition.                       | Current draft Section IV-C-2; cite Kipf & Welling [9], Hamilton et al. [27], Gilmer et al. [28] depending on exact layer formulation. | Do not copy an external equation unless it is the actual layer used. If the implementation has a custom update, derive the equation from the code. |
| Attention                           | Optional architecture.                         | Current draft Section IV-C-2; cite Velickovic et al. [10].                                                                            | Only write a mathematical attention equation if the implemented attention layer is explicitly characterized and ablated.                           |
| Fiber/FSO physical equations        | Channel model.                                 | Current draft states the implementation and points to PHYSICS_EVIDENCE.md; detailed code/model provenance is outside the paper text.  | Extract the exact implemented equations from the physics evidence/code. Do not invent a textbook equation and claim it is the implementation.      |
| Johann QBER model                   | Only if adopting Johann's absolute-rate model. | Johann paper p. 384, Eq. (1).                                                                                                         | Use only if you actually port that model. Cite Johann and preserve parameter definitions.                                                          |
| Johann post-processing factor       | Absolute key generation extension.             | Johann p. 384, Eq. (2).                                                                                                               | Use with the same caveat: only after adopting their model and assumptions.                                                                         |
| Johann Nsec                         | Key-block generation extension.                | Johann p. 384, Eq. (3).                                                                                                               | Useful for converting absolute key-generation model into 256-bit key blocks if you reproduce that framework.                                       |
| Weight heuristic $W_e = \exp(aE_e)$ | Classical state-aware comparator.              | Current draft Section IV-D-1, Eq./definition text; attributed to [4].                                                                 | Implement only with QKP state. Cite Johann [4]. A rate-aware modification is a new method and must be labeled as such.                             |
| ILP objective/constraints           | Global demand-level comparator.                | Current draft Section V, Eqs. (5)–(8), PDF p. 6; formulation is mathematical only at present.                                         | Do not report performance until demand/capacity simulation exists. Use as global reference, not next-hop baseline.                                 |

**Equation provenance rule: every equation in the final paper should be traceable to one of three sources: (A) already implemented/current draft; (B) a cited external reference whose model is actually adopted; or (C) a new equation derived from the proposed method and supported by the implementation. Never insert an equation merely because it is standard if the simulator does something different.**

## 8. REFERENCE MAP — WHAT TO CITE FOR WHAT

| Ref.                                            | Use it for                                                                                                                                      | Do not use it for                                                                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| [4] Johann et al., JOCN 2024                    | Trusted-node routing, QKP architecture, demand simulation, k-shortest paths, weight optimization, blocking, load balancing, management traffic. | Do not claim their absolute key-rate model is your implemented model unless ported.   |
| [8] Schulman et al., PPO                        | PPO algorithm and clipped objective.                                                                                                            | Not a QKD routing reference.                                                          |
| [9] Kipf & Welling, GCN                         | Graph convolution/message aggregation background.                                                                                               | Not necessarily the exact architecture unless your layer matches.                     |
| [10] Velickovic et al., GAT                     | Attention mechanism background.                                                                                                                 | Do not imply attention improved performance without your ablation.                    |
| [11] Bennett & Brassard                         | BB84 background.                                                                                                                                | Not a finite-key certificate for your hybrid network.                                 |
| [12], [13] Hwang; Lo-Ma-Chen                    | Decoy-state background.                                                                                                                         | Do not claim the current proxy implements their full security analysis.               |
| [14] Gottesman et al.                           | Security with imperfect devices.                                                                                                                | Use as security background, not as validation of the simulator.                       |
| [15] Tomamichel et al.                          | Finite-key analysis foundations.                                                                                                                | Use if discussing finite-size security limitations or future rigorous estimator.      |
| [16] Lim et al.                                 | Practical decoy-state security bounds.                                                                                                          | Same boundary: background unless actually implemented.                                |
| [17] ITU-T Y.3803                               | QKD key-management/network architecture context.                                                                                                | Not a source of measured performance.                                                 |
| [18] ITU-R P.1814-1                             | Terrestrial FSO link-design/prediction context.                                                                                                 | Not a calibration dataset for the current Indian topology.                            |
| [19] Krzic et al., npj Quantum Information 2023 | Metropolitan free-space quantum network context.                                                                                                | Not direct validation of your weather assumptions.                                    |
| [20] Dijkstra                                   | Classical shortest-path baseline.                                                                                                               | Not a QKD-specific resource-aware algorithm.                                          |
| [21] Roik et al.                                | Related reinforcement-learning routing in quantum networks.                                                                                     | Use for positioning, not direct apples-to-apples comparison.                          |
| [22] Amer et al.                                | Dynamic QKD routing context.                                                                                                                    | Use for related-work positioning and future demand-aware extension.                   |
| [23] Bellman                                    | Dynamic programming background if needed.                                                                                                       | Not necessary for the core paper unless discussing classical sequential optimization. |
| [24] Schulman et al., GAE                       | Generalized advantage estimation.                                                                                                               | Use only in RL methodology.                                                           |

| Ref.                                                      | Use it for                                 | Do not use it for                                          |
|-----------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|
| [25], [26]<br>POMDP/RL<br>foundations                     | Partial observability and RL formulation.  | Not evidence that the specific QKD environment is correct. |
| [27], [28],<br>[29] GNN/MP<br>NN/attention<br>foundations | Graph representation and attention theory. | Use only where the implemented architecture corresponds.   |
| [30] OSM                                                  | Map-data attribution.                      | Do not describe road corridors as verified fiber routes.   |

## 9. FIGURE PLAN — EXACTLY WHAT TO DRAW

| Fig. | Priority    | Figure                                 | How to generate it                                                                                                                               | Message/caption                                                       |
|------|-------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| F1   | KEEP/UPDATE | Normalized rate proxy vs link distance | Sweep fiber/FSO distance under fixed scenarios; shade only sampled simulator variation.                                                          | “Modeled normalized rate proxy; not measured link performance.”       |
| F2   | KEEP        | TN vs STN conceptual architecture      | Existing conceptual schematic; retain clear boundary between conventional TN relay and STN abstraction.                                          | “Security architecture context; STN proof is not implemented.”        |
| F3   | KEEP/REFINE | Hybrid routing architecture            | Draw topology → environment → GNN/PPO → action mask → route/key-pool state. Use current Fig. 3 concept.                                          | Show what is implemented vs conceptual OTP/XOR/key-management plane.  |
| F4   | KEEP/UPDATE | Indian corridor graph                  | Plot 2,370 nodes/2,653 links; distinguish fiber and candidate FSO; mark Mumbai→Kolkata example.                                                  | Explicitly label road geometry/FSO visibility as proxies/assumptions. |
| F5   | KEEP        | Channel-to-routing coupling            | Use current pipeline: scenario → channel model → link estimates → rate model → policy mask/features. Mark non-implemented models.                | This is the physics-to-policy contribution.                           |
| F6   | KEEP/UPDATE | Single-route evaluation flow           | Load config → reset → policy/baseline → feasibility → update → terminate → aggregate seeds.                                                      | Update once corrected baselines are regenerated.                      |
| F7   | KEEP/UPDATE | GNN-PPO architecture                   | Graph features → 3 message-passing layers → candidate scoring → masked actor; critic branch. Show LSTM only if comparator is actually evaluated. | Do not visually imply attention benefit unless ablated.               |
| F8   | NEW P0      | Physics validation panel               | Fiber loss vs distance; detection vs loss; QBER vs noise; rate proxy vs QBER; FSO availability/correlation.                                      | Use analytical/limiting-case checks, not decorative curves.           |
| F9   | NEW P0      | Policy physical sensitivity            | Scatter/curve of P(select B) vs $\Delta R$ , plus matched QBER/outage panels.                                                                    | Directly addresses the current zero-FSO-selection diagnostic.         |
| F10  | NEW P0/P1   | Robustness matrix                      | Rows = test condition; columns = success, hops, QBER validity, FSO usage.                                                                        | Use fixed seeds and report distributions.                             |
| F11  | NEW P1      | Training stability                     | Five-seed learning curves with mean + spread; optionally individual faint traces.                                                                | Demonstrate whether behavior is reproducible.                         |
| F12  | NEW P1      | Inference scaling                      | p50/p95/p99 latency vs graph size; CPU/GPU comparison if relevant.                                                                               | Separate model-forward time from route-search/pre processing.         |
| F13  | NEW P2      | Demand/QKP state machine               | Demand → route → consume → refill → next interval.                                                                                               | Central figure for journal extension.                                 |
| F14  | NEW P2      | Blocking vs traffic scaling            | Traffic scaling factor on x-axis; blocking probability on y-axis; methods as series.                                                             | This is the closest direct network-level comparison to Johann.        |
| F15  | NEW P2      | QKP utilization                        | Time-series and/or distribution of QKP filling levels; optionally regional focus.                                                                | Shows load balancing and depletion/recovery.                          |

| Fig. | Priority | Figure             | How to generate it                                      | Message/caption                                          |
|------|----------|--------------------|---------------------------------------------------------|----------------------------------------------------------|
| F16  | NEW P2   | Management traffic | Controller requests/bytes vs traffic factor and method. | Must define what constitutes a management request first. |

## 10. TABLE PLAN — WHAT SHOULD BE IN THE PAPER

| Table | Purpose                      | Columns / content                                                                               | Status                                                                              |
|-------|------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| T1    | Link-model provenance        | Parameter, value, unit, source/evidence class, calibrated vs assumed.                           | <b>DONE conceptually; update provenance.</b>                                        |
| T2    | Observation/action features  | Node features, edge features, action mask conditions.                                           | <b>DONE.</b>                                                                        |
| T3    | Archived evaluation scenario | Endpoint, season/time, training budget, hidden size, seeds, QBER threshold, outage probability. | <b>DONE; replace “10 seeds” wording if it actually means evaluation seeds only.</b> |
| T4    | Baseline definition          | Method, information available, feasibility rule, objective, computational role.                 | <b>TODO; essential after corrected baselines.</b>                                   |
| T5    | P0 result summary            | Success, hops, route length, rate utility, invalid actions, revisits, FSO usage across methods. | <b>TODO.</b>                                                                        |
| T6    | Ablation summary             | Architecture/features/reward variants and metric deltas.                                        | <b>P1.</b>                                                                          |
| T7    | Inference benchmark          | Parameters, memory, p50/p95/p99, CPU/GPU, graph size.                                           | <b>P1.</b>                                                                          |
| T8    | Journal network KPIs         | Blocking, served demand, QKP utilization, depletion events, management requests.                | <b>P2.</b>                                                                          |

## 11. PAPER WRITING MAP — WHERE EACH RESULT GOES

| Paper section                      | Write this                                                                                                                                                                                                                                                          | Use these existing sources/evidence                                                                     | Add before finalizing                                           |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| I. Introduction                    | Problem: trusted-node routing must jointly account for link quality, changing channel state, trust constraints and stored-key availability. State that the current contribution is a route-level hybrid fiber/FSO controller, not a complete network-service study. | Current draft pp.1-2; Johann [4].                                                                       | Make contribution claims match actual P0 evidence.              |
| II-A. QKD background               | BB84, trusted-node relay, STN distinction, and finite-key boundary.                                                                                                                                                                                                 | Current draft p.2; refs [11]–[17].                                                                      | Keep security claims strictly scoped.                           |
| II-B. Link/rate model              | Fiber/FSO assumptions, QBER proxy, normalized rate proxy. Equation (1) exactly from current draft.                                                                                                                                                                  | Current draft p.4; PHYSICS_EVIDENCE.md/code for implementation details; refs [18],[19] for FSO context. | Add analytical validation results.                              |
| III-A. Architecture/topology       | Four simulator components, 7 hubs, topology counts, proxy limitations.                                                                                                                                                                                              | Current draft p.3; OSM [30].                                                                            | Updated topology figure.                                        |
| III-B. Environment                 | Observation, action mask, QBER threshold, route history, pool abstraction.                                                                                                                                                                                          | Current draft p.4.                                                                                      | Correct any feature descriptions after code audit.              |
| IV-A. Routing formulation          | Graph/POMDP notation and discounted objective Eq. (2).                                                                                                                                                                                                              | Current draft p.5; refs [25],[26].                                                                      | Clarify exactly what is partially observed.                     |
| IV-B. GNN/action mask              | Message passing, masked policy Eq. (3), architecture figure.                                                                                                                                                                                                        | Current draft p.5; refs [9],[10],[27],[28].                                                             | Add ablation evidence before claiming architectural benefit.    |
| IV-C. PPO                          | Clipped objective Eq. (4); GAE/training details.                                                                                                                                                                                                                    | Current draft p.5; refs [8],[24].                                                                       | Report protected hyperparameters and seed protocol.             |
| IV-D. Classical comparators        | BFS/Dijkstra/Max-SKR. Explain why Johann k-shortest/QKP algorithms are not apples-to-apples yet.                                                                                                                                                                    | Current draft p.5; Johann [4]; Dijkstra [20].                                                           | Regenerate corrected baseline results.                          |
| V. Results                         | P0 physics validation → baseline route results → policy mechanism → robustness/generalization → inference.                                                                                                                                                          | Archived current results + new P0/P1 outputs.                                                           | Do not use old invalid baseline rows.                           |
| VI. Discussion                     | Explain what the policy learned, what it failed to learn, information/computation tradeoffs, and simulator limitations.                                                                                                                                             | Controlled link-choice diagnostic + validation evidence.                                                | Use measured language; no superiority claim from one condition. |
| VII. Journal extension/future work | Absolute QKP, dynamic demand, refill, blocking, load balancing, management traffic, prediction and ILP.                                                                                                                                                             | Johann pp.384–390; current draft limitations.                                                           | Only claim implementation after actually building it.           |

## 12. BASE-PAPER GAP MAP — WHAT TO IMPORT, WHAT NOT TO IMPORT

| Johann component                  | Current state                                                               | Decision                      | When to implement                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------|
| Dynamic demand matrices           | <b>Absent</b>                                                               | IMPORT                        | P2 journal layer.                                                                            |
| 10,000 × 256-bit initial QKPs     | <b>Absent</b>                                                               | IMPORT AS REPRODUCTION OPTION | Only with explicit citation/assumption; do not present as measured hardware state.           |
| Key consumption on every path QKP | <b>Absent</b>                                                               | IMPORT                        | P2.                                                                                          |
| 60-s refill/generation loop       | <b>Absent</b>                                                               | IMPORT/ADAPT                  | Use exact interval only if reproducing Johann; otherwise define a justified interval.        |
| Blocking probability              | <b>Absent</b>                                                               | IMPORT                        | P2.                                                                                          |
| QKP load balancing/utilization    | <b>Absent</b>                                                               | IMPORT                        | P2.                                                                                          |
| Management traffic                | <b>Absent</b>                                                               | IMPORT CONCEPT                | Define your own controller/query semantics before counting.                                  |
| K=5 loopless k-shortest paths     | <b>Absent</b>                                                               | OPTIONAL IMPORT               | Useful for demand-level classical comparator; not necessary for immediate route-level paper. |
| Perfect prediction                | <b>Absent</b>                                                               | IMPORT LATER                  | Only after demand history exists                                                             |
| LSTM prediction                   | <b>Current LSTM is graph-blind node-array encoder, not demand predictor</b> | DO NOT CONFUSE                | Replace/augment with temporal-demand LSTM in P2 if needed.                                   |
| Weight optimization               | <b>Current draft gives the heuristic definition but does not measure it</b> | IMPORT LATER                  | Needs absolute QKP state and demand loop.                                                    |
| ILP                               | <b>Current draft has Eqs. (5)-(8), no measured result</b>                   | KEEP AS FORMULATION           | Implement only in P2 after resource/demand layer.                                            |

**Important distinction: Johann's LSTM predicts future traffic demand. The current manuscript's LSTMActorCritic is a graph-blind encoder over an ordered node-feature array. They are different objects and must not be described as the same baseline.**

## 13. AGENT EXECUTION PROTOCOL — HOW TO WORK WITHOUT LOSING THE SCIENTIFIC THREAD

- Step 1 — Freeze provenance. Every run gets a versioned config, code commit/hash, random seed, environment version, topology version, and output directory.
- Step 2 — Never overwrite archived evidence. Keep the 25.6-hop run and 384-pair diagnostic as archived artifacts; label them preliminary.
- Step 3 — Fix the environment before comparing methods. The selector must use the observation action ordering and valid-action mask. No environment fallback should silently replace a baseline action.
- Step 4 — Separate training from evaluation. Hyperparameters are chosen on training/validation scenarios; final test seeds and held-out conditions are not used for tuning.
- Step 5 — Separate model error from policy error. First validate attenuation/QBER/rate/FSO statistics. Then test whether the policy responds correctly to those quantities.
- Step 6 — One scientific question per experiment. Do not change endpoint, weather, reward, topology and model architecture simultaneously if the goal is causal interpretation.
- Step 7 — Every plot gets a data provenance sentence. State whether points are measured, analytically calculated, simulated from assumptions, or derived from an archived policy run.
- Step 8 — Every new equation gets a provenance tag. Current draft equation, adopted literature equation, or newly derived method equation.
- Step 9 — Do not manufacture a missing comparison. If Gymnasium/runtime/dependency prevents regeneration, report the blocker and keep the old result out of the comparative table.
- Step 10 — Update the paper only after the result artifact exists. The writing agent should not draft a result claim from an intended experiment.

### Recommended output directory

```
research_runs/  
├─ 00_archived/  
├─ 01_baseline_corrected/  
├─ 02_physics_validation/  
├─ 03_link_choice/  
├─ 04_qber_fso_stress/  
├─ 05_multiseed/  
├─ 06_generalization/  
├─ 07_ablations/  
├─ 08_inference/  
└─ 09_journal_network_layer/
```

Each directory should contain: config.yaml, run\_manifest.json, raw\_metrics.csv, summary.csv, figures/, and a short README stating the hypothesis, controlled variables, result, and paper section.

## 14. CLAIMS CONTROL — WHAT THE PAPER MAY SAY

| Evidence state                  | Allowed wording                                                                         | Avoid                                                            |
|---------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Current simulator model         | "We model..." / "The simulator computes..."                                             | "Measured..." / "validated experimentally..."                    |
| 80-km fiber / 10-km FSO outputs | "Under the configured parameters..."                                                    | "The link achieves..."                                           |
| 10/10 GNN archived run          | "In the archived Mumbai-Kolkata condition, the policy completed 10/10 episodes."        | "The method is robust/optimal."                                  |
| 25.6 hops                       | "Mean 25.6 hops in the archived run; the independently checked topology minimum is 24." | "Near-optimal" unless supported by a proper route-quality study. |
| 384-pair link diagnostic        | "The policy selected the higher-proxy FSO candidate in 0/384 controlled comparisons."   | "The GNN understands/does not understand physics" — too broad.   |
| P0 multi-seed result            | "Across five independent training seeds..."                                             | Single-run generalization claims.                                |
| Journal blocking result         | "Under the specified demand/QKP model..."                                               | General claims about deployed QKD network throughput.            |
| ILP                             | "We formulate an ILP comparator..."                                                     | "ILP outperforms the learned policy" before execution.           |
| Field relevance                 | "Provides a simulation framework for..."                                                | "Demonstrates real-world network performance."                   |

## 15. MASTER PRIORITY QUEUE

| Priority | Do this                                   | Definition of done                                                                        | Paper impact                              |
|----------|-------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------|
| P0.1     | Regenerate corrected baselines            | All methods use same mask, endpoint, seeds, termination and metrics; raw CSV archived.    | Unlocks valid Results comparison.         |
| P0.2     | Validate physics/noise                    | Analytical trends + independent calculations + FSO stochastic checks pass.                | Makes channel model defensible.           |
| P0.3     | Run controlled link-choice/QBER/FSO tests | Policy probability and selection behavior quantified against controlled physical changes. | Explains mechanism/current weakness.      |
| P0.4     | Run 5 independent training seeds          | Fixed held-out evaluation; mean/spread reported.                                          | Removes single-run limitation             |
| P1.1     | Endpoint/season/diurnal generalization    | Held-out conditions complete with confidence intervals.                                   | Makes robustness claim possible.          |
| P1.2     | Feature/architecture/reward ablations     | Same training budget and seed distribution; effects quantified.                           | Explains what drives policy.              |
| P1.3     | Inference benchmark                       | p50/p95/p99, memory, CPU/GPU and scaling reported.                                        | Adds implementation/deployment relevance. |
| P1.4     | Small-graph oracle                        | Exact feasible-path reference and regret computed.                                        | Adds route-quality reference.             |
| P2.1     | Absolute QKP model                        | Capacity, initial inventory, generation and consumption units defined.                    | Unlocks network-service metrics.          |
| P2.2     | Dynamic demand loop                       | Demand → route → consume → refill works over time.                                        | Turns route study into network study.     |
| P2.3     | Blocking/load balancing                   | Traffic scaling + QKP utilization + recovery measured.                                    | Direct Johann comparison.                 |
| P2.4     | Management traffic                        | Controller query semantics and overhead measured.                                         | Information/security tradeoff             |
| P2.5     | Weight/LSTM/perfect prediction            | Classical global-state and temporal-demand comparators implemented.                       | Journal baseline family.                  |
| P2.6     | ILP                                       | Demand-level capacity model solved at defined scale; runtime recorded.                    | Global reference/oracle.                  |
| P3       | Measured traces / HIL                     | Trace provenance + model-vs-measurement error + policy transfer reported.                 | External validation.                      |

**Recommended immediate stopping point: finish P0 and P1 before building the full Johann-style demand/QKP layer. The immediate paper then has a complete scientific story: noise-aware hybrid network model → constrained routing problem → GNN-PPO controller → corrected classical comparison → policy mechanism → robustness/generalization → inference cost → explicit limitations.**

Journal story after P2: extend that same controller into a sustained network-service environment where routing decisions consume and replenish finite QKD resources under dynamic demand, enabling blocking, load balancing, management overhead, prediction, and global optimization comparisons.

## 16. SOURCE LOCATOR — QUICK REFERENCE FOR THE AGENT

| Source                            | Where    | Use                                                                                                                    |
|-----------------------------------|----------|------------------------------------------------------------------------------------------------------------------------|
| Current QKD_IEEE_Paper.pdf        | p. 1     | Abstract, contribution boundary, current evidence scope.                                                               |
| Current QKD_IEEE_Paper.pdf        | p. 2     | QKD/TN/STN background, finite-key boundary, rate interpretation.                                                       |
| Current QKD_IEEE_Paper.pdf        | p. 3     | Architecture, topology, parameter table, proxy-network assumptions.                                                    |
| Current QKD_IEEE_Paper.pdf        | p. 4     | FSO model, Eq. (1), observation features, action mask, archived setup.                                                 |
| Current QKD_IEEE_Paper.pdf        | p. 5     | Simulation flow, Eq. (2)–(4), GNN/action mask, PPO, weight heuristic.                                                  |
| Current QKD_IEEE_Paper.pdf        | p. 6     | ILP Eqs. (5)–(8), preliminary 10/10 and 25.6-hop result, key-store limitation.                                         |
| Current QKD_IEEE_Paper.pdf        | p. 7     | Physics evidence, controlled link-choice diagnostic, scope/validation limitations.                                     |
| Current QKD_IEEE_Paper.pdf        | p. 8     | References [14]–[30].                                                                                                  |
| Johann et al. JOCN 2024           | p. 384   | Absolute QKD key-generation model: QBER, post-processing factor, secret-key block calculation.                         |
| Johann et al. JOCN 2024           | p. 385   | Central controller, QKP architecture, security/management-information tradeoff.                                        |
| Johann et al. JOCN 2024           | p. 386   | Dynamic demand matrix processing, key consumption, refill, K=5 loopless shortest paths.                                |
| Johann et al. JOCN 2024           | p. 387   | Perfect prediction and LSTM demand prediction.                                                                         |
| Johann et al. JOCN 2024           | p. 389   | Blocking probability and QKP utilization/load balancing.                                                               |
| Johann et al. JOCN 2024           | p. 390   | Management traffic and overall tradeoffs.                                                                              |
| Uploaded QSMS Presentation Script | pp. 1–11 | Visual reference only: DejaVu Sans, burgundy #9E1735, pale #F6F1F2, white background, restrained tables and sectioning |

**This roadmap is designed so another research agent can resume work without reconstructing the project history: start at Section 2 for current state, Section 3 for missing dependencies, Section 4 for P0, Section 7 for equation provenance, Section 9 for figure recipes, and Section 11 for exact paper placement.**

# RESEARCH ROADMAP

## STATUS UPDATE - 2026-10-02

**How to read this update.** This addendum supplements the original 22-page roadmap. It records work completed in the repository, generated evidence, and remaining dependencies. Earlier archived results remain historical records and are not silently replaced. All current outputs are simulator-derived; none is field validation or a QKD security proof.

### 1. Work completed in this update

| Workstream                               | Change / result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Remarks and evidence                                                                                                                                                           |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corrected route comparison               | Baseline selectors use the observation's feasible-action ordering; all methods use the same season and channel seeds. Per-episode output records action validity, no-feasible-action states, fallbacks, route, hops, revisits, and link validity.                                                                                                                                                                                                                                                                | The comparison output is identified below. No-feasible-action states are separated from invalid policy actions.                                                                |
| Physics consistency                      | Independent formula transcription covered 201 fiber distances from 0 to 200 km. Maximum absolute difference: QBER 0; normalized proxy 0. Transmission and proxy decrease; QBER does not decrease.                                                                                                                                                                                                                                                                                                                | Internal consistency only. It does not validate assumptions against measurements or establish security.                                                                        |
| FSO stochastic checks                    | All 16 configured season/distance anchors sampled at 2,000 trials each; maximum absolute availability deviation 0.0125 (1.22 target standard errors). Log-Cn2 AR(1): configured 0.85, sampled 0.8522 over 20,000 steps.                                                                                                                                                                                                                                                                                          | These are model targets and samples, not validation against Indian weather observations.                                                                                       |
| Link-feature sensitivity                 | The controlled diagnostic includes a proxy-rate ratio sweep, a feasible QBER-margin sweep, and an outage-mask counterfactual with route context held fixed. The FSO action probability rose from 0.311 to 0.638 as its proxy-rate ratio rose from 0.25x to 4x; at QBER 0.111 it was masked. In a separate 32-sample FSO turbulence counterfactual, a 100x Cn2 multiplier lowered median proxy from 0.202 to 0.183 and mean FSO action probability from 0.749 to 0.740; FSO remained selected in 100% of samples. | A policy-input diagnostic does not establish link-measurement validity or route-level performance.                                                                             |
| Five-seed stability campaign             | 2/5 independent policies have final evaluations. Seed 20261004 is the latest run without a final summary. Completed seed 20261002: 93.8% success, 69.80 mean hops among successful episodes.                                                                                                                                                                                                                                                                                                                     | Inter-seed spread and confidence summaries are withheld until all planned seeds finish. A training-seed interval is distinct from an episode-level interval.                   |
| Exact toy-graph route oracle             | Exhaustive simple-path reward search covered 20 graph/channel cases with a feasible oracle in 20. The policy succeeded in 90%; it exactly matched oracle reward in 55% of all cases and 61% of successful cases.                                                                                                                                                                                                                                                                                                 | Five generated 7-node graphs; episode link states frozen with 5% per-edge outage. This exact result is limited to the toy static-channel protocol, not the full dynamic graph. |
| Paired endpoint / season / time transfer | Across 108 GNN episodes, success was 99/108 (91.7%; Wilson 95% CI 84.9%-95.6%); successful mean hops 31.94. BFS-hop success was 108/108 (100.0%), with 15.40 successful mean hops. GNN did not outperform BFS on success or hops.                                                                                                                                                                                                                                                                                | Six ordered pairs, four seasons and selected day/night times, paired seeds. BC endpoint randomization means these are not strict end-to-end endpoint holdouts.                 |
| Repeated inference profile               | 1 thread(s): p50/p95/p99 26.52/70.37/94.68 ms; 4 thread(s): p50/p95/p99 22.13/45.12/64.58 ms; 8 thread(s): p50/p95/p99 12.76/16.89/22.58 ms. Process RSS after run 348.4 MiB; model parameters 0.5 MiB.                                                                                                                                                                                                                                                                                                          | 42 ordered pairs, deterministic single-decision calls, encoder cache disabled. One host/topology; graph-size scaling and deployment profiling remain open.                     |
| Matched-budget ablation campaign         | 0/5 conditions complete at the shared 50-epoch budget.                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Single training seed; progress is not a completed comparative result.                                                                                                          |

### 2. Paired route comparison

| Policy  | Success                             | Successful mean hops | Invalid actions | No-valid-action states | Revisits |
|---------|-------------------------------------|----------------------|-----------------|------------------------|----------|
| GNN-PPO | 16/16 (100.0%; 95% CI 80.6%-100.0%) | 28.25                | 0               | 0                      | 2        |

| Policy      | Success                             | Successful mean hops | Invalid actions | No-valid-action states | Revisits |
|-------------|-------------------------------------|----------------------|-----------------|------------------------|----------|
| Random      | 0/16 (0.0%; 95% CI 0.0%-19.4%)      | n/a                  | 0               | 8                      | 2459     |
| Dijkstra-km | 4/16 (25.0%; 95% CI 10.2%-49.5%)    | 233.00               | 0               | 0                      | 172      |
| BFS-hop     | 16/16 (100.0%; 95% CI 80.6%-100.0%) | 28.25                | 0               | 0                      | 3        |
| Max-SKR     | 0/16 (0.0%; 95% CI 0.0%-19.4%)      | n/a                  | 0               | 5                      | 984      |

Conditions: Delhi to Chennai; 4 shared environment seeds per each of 4 seasons; QBER/rate action mask; 400 step limit. Scope is single-route simulator behavior, not traffic blocking, network-wide key-pool balance, or delivered service capacity.

## Checkpoint transfer across city pairs

The same 50-epoch checkpoint completed 31/42 ordered city-pair episodes (73.8%) under the archived monsoon-night scenario. Median successful route length was 20 hops; localized decision latency was 3.83 ms median and 4.34 ms p95 on this host. This is one stochastic sample per ordered pair, not a held-out endpoint protocol or a matched baseline comparison.

### 3. Roadmap status after this work

| Priority                            | Status                     | Evidence / remaining work                                                                                                                                                                                                                                                    |
|-------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0-1 corrected route comparison     | DONE - route-level         | Paired environment seeds, shared feasibility rules, and action/fallback accounting. The separate demand-driven QKP comparison remains out of scope.                                                                                                                          |
| P0-2 physics checks                 | DONE - internal only       | Fiber formula consistency, configured FSO anchor sampling, and temporal correlation. No matched device/channel calibration or measured traces.                                                                                                                               |
| P0-3 link sensitivity               | DONE - diagnostic          | Rate-ratio, QBER-margin, and hard outage-mask tests vary simulator policy inputs only.                                                                                                                                                                                       |
| P0-4 QBER stress                    | PARTIAL                    | Controlled action-probability sweep covers QBER 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.109 and 0.111. The out-of-threshold candidate is masked. Route success, entropy, and preference curves across endpoint/weather conditions remain.                                      |
| P0-5 FSO stress                     | PARTIAL                    | A 0.01x-100x Cn2 input sweep, season/hour quality samples, configured marginal availability, and an outage-mask counterfactual exist. Action-level response only: no route outcomes; pointing/background parameters are fixed and outage-persistence bursts are unsupported. |
| P0-6 five training seeds            | IN PROGRESS - 2/5 complete | Five independent full-budget training runs are not present. Evaluation episodes for one checkpoint do not substitute for training-seed replication.                                                                                                                          |
| P1 endpoint / weather holdouts      | PARTIAL - transfer matrix  | Paired across-pair and season/time matrix is archived (108 episodes per policy) with Wilson intervals. GNN success was 91.7% versus BFS 100.0%; successful mean hops were 31.94 versus 15.40. BC endpoint randomization prevents a strict endpoint-holdout claim.            |
| P1 architecture / feature ablations | IN PROGRESS (0/5)          | Matched 50-epoch baseline, geographic-feature, no-SKR-reward, no-DropEdge, and width-32 runs are underway.                                                                                                                                                                   |
| P1 inference cost                   | PARTIAL - profiled         | Repeated per-decision CPU latency, p50/p95/p99, thread counts, host details, and process/model memory are archived across 42 ordered pairs. This remains a one-host, one-topology simulator profile; no graph-size scaling or deployment claim.                              |
| P1 exact small-graph oracle         | DONE - toy scope           | Exhaustive simple-path reward oracle and policy regret archived for generated 7-node graphs with frozen episode channels. Does not establish optimality on the full dynamic network.                                                                                         |
| P2 network-service study            | BLOCKED BY MODEL SCOPE     | Requires absolute key generation and QKP capacity, time-indexed demand, consumption/refill, blocking, load balance, controller message semantics, prediction baselines, and executable demand-level ILP.                                                                     |
| P3 external validation              | BLOCKED BY DATA / HARDWARE | Requires matched device/channel measurements or hardware-in-the-loop and separate model-error analysis.                                                                                                                                                                      |

### 4. Interpretation and next work

The corrected comparison supports a route-level simulator statement only. Report success, successful-route hops, revisits, action-mask violations, and route outcomes; do not infer network capacity or superiority from reward alone. Preserve baseline failures as observed outcomes and state each baseline definition.

Finish the remaining QBER/FSO stress outcomes and the five-seed campaign, then run held-out endpoint/weather experiments and matched-budget ablations. The Johann-style demand/QKP study is a separate extension and depends on absolute key-resource units and a time-indexed demand model. Do not present Johann's BB84 equations as implemented until that model is adopted and validated.

Regression check: 41 tests passed, 0 failed in 68.8 s; 1 warning(s) recorded. Details: `test_results_20261002.json`.

**Provenance rule:** Label every numeric result as measured, independently calculated, model-sampled, or derived from a policy run. Internal consistency checks do not upgrade an assumption to calibrated physics.